

CSE 400: Stochastic Processes

Lecture 26: SSS, WSS, and Ergodic Processes Scribe Notes

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1 Introduction

A **stochastic process** is a collection of random variables indexed by time:

$$X(t), \quad t \in T$$

Each outcome produces a **realization** (sample function).

1.1 Key Concepts

- **Random Variable:** $X(t)$ at fixed t
- **Realization:** Function of time for fixed outcome
- **Ensemble:** Collection of all realizations

2 Characterization of Stochastic Process

2.1 Distribution Functions

- First-order PDF: $f_X(x, t)$
- Second-order PDF: $f_X(x_1, t_1; x_2, t_2)$
- General n -th order PDF

3 Statistical Moments

3.1 Mean Function

$$\mu_X(t) = E[X(t)]$$

3.2 Autocorrelation Function

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$$

3.3 Covariance Function

$$C_X(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(X(t_2) - \mu_X(t_2))]$$

Relation:

$$C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$$

4 Properties of Autocorrelation

- Symmetry:

$$R_X(t_1, t_2) = R_X(t_2, t_1)$$

- Maximum at origin:

$$R_X(t, t) \geq R_X(t_1, t_2)$$

- Non-negative:

$$R_X(t, t) \geq 0$$

5 Positive Definite Function

A function R_X is **positive definite** if:

$$\sum_{i=1}^n \sum_{j=1}^n a_i a_j R_X(t_i, t_j) \geq 0$$

5.1 Matrix Representation

$$\mathbf{R} = \begin{bmatrix} R(t_1, t_1) & \cdots & R(t_1, t_n) \\ \vdots & \ddots & \vdots \\ R(t_n, t_1) & \cdots & R(t_n, t_n) \end{bmatrix}$$

5.2 Condition

$$\mathbf{a}^T \mathbf{R} \mathbf{a} \geq 0$$

6 Stationary Processes

6.1 Strict Sense Stationary (SSS)

A process is SSS if all statistical properties are invariant under time shift:

$$f_X(x_1, t_1; \dots; x_n, t_n) = f_X(x_1, t_1 + \tau; \dots; x_n, t_n + \tau)$$

6.2 Wide Sense Stationary (WSS)

A process is WSS if:

- Mean is constant:

$$\mu_X(t) = \mu$$

- Autocorrelation depends only on time difference:

$$R_X(t_1, t_2) = R_X(\tau), \quad \tau = t_2 - t_1$$

6.3 SSS vs WSS

- SSS $\Rightarrow$ WSS
- WSS $\nRightarrow$ SSS (in general)

6.4 Gaussian Process Special Case

For Gaussian processes:

$$\text{WSS} \Rightarrow \text{SSS}$$

7 Example: White Noise Process

- Mean:

$$\mu = 0$$

- Autocorrelation:

$$R_X(\tau) = \sigma^2 \delta(\tau)$$

8 Ergodic Process

A process is **ergodic** if time averages equal ensemble averages.

8.1 Time Average

$$\overline{X} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} X(t) dt$$

8.2 Ergodicity Condition

$$\overline{X} = E[X(t)]$$

9 Example: Mean and Autocorrelation Derivation

9.1 Mean

$$\mu_X(t) = E[X(t)]$$

9.2 Autocorrelation

$$R_X(t_1, t_2) = E[X(t_1)X(t_2)]$$

Steps:

1. Express process
2. Compute expectation
3. Use independence (if applicable)

10 Important Mathematical Tools

- Expectation operator linearity
- Independence:

$$E[XY] = E[X]E[Y]$$

- Delta function properties

11 Exam-Oriented Summary

11.1 Key Definitions

- Stochastic Process = Collection of RVs
- Mean: $E[X(t)]$
- Autocorrelation: $E[X(t_1)X(t_2)]$
- Covariance: $R - \mu^2$

11.2 Stationarity

- SSS: Full distribution invariant
- WSS: Mean constant + autocorrelation depends only on τ

11.3 Important Results

- SSS $\Rightarrow$ WSS
- Gaussian: WSS $\Rightarrow$ SSS

11.4 Autocorrelation Properties

- Symmetric
- Positive definite
- Maximum at origin

11.5 Ergodicity

- Time average = Ensemble average

11.6 Must-Know Examples

- White noise
- Mean and autocorrelation derivation

12 Final Remarks

This lecture builds the foundation for:

- Signal processing
- Random signal analysis
- Communication systems